\date{2025-2026}

Robotic Engineer Portofolio

Master 2 Advanced

Systems and Robotics SAR

Cursus Master en Ingénierie CMI

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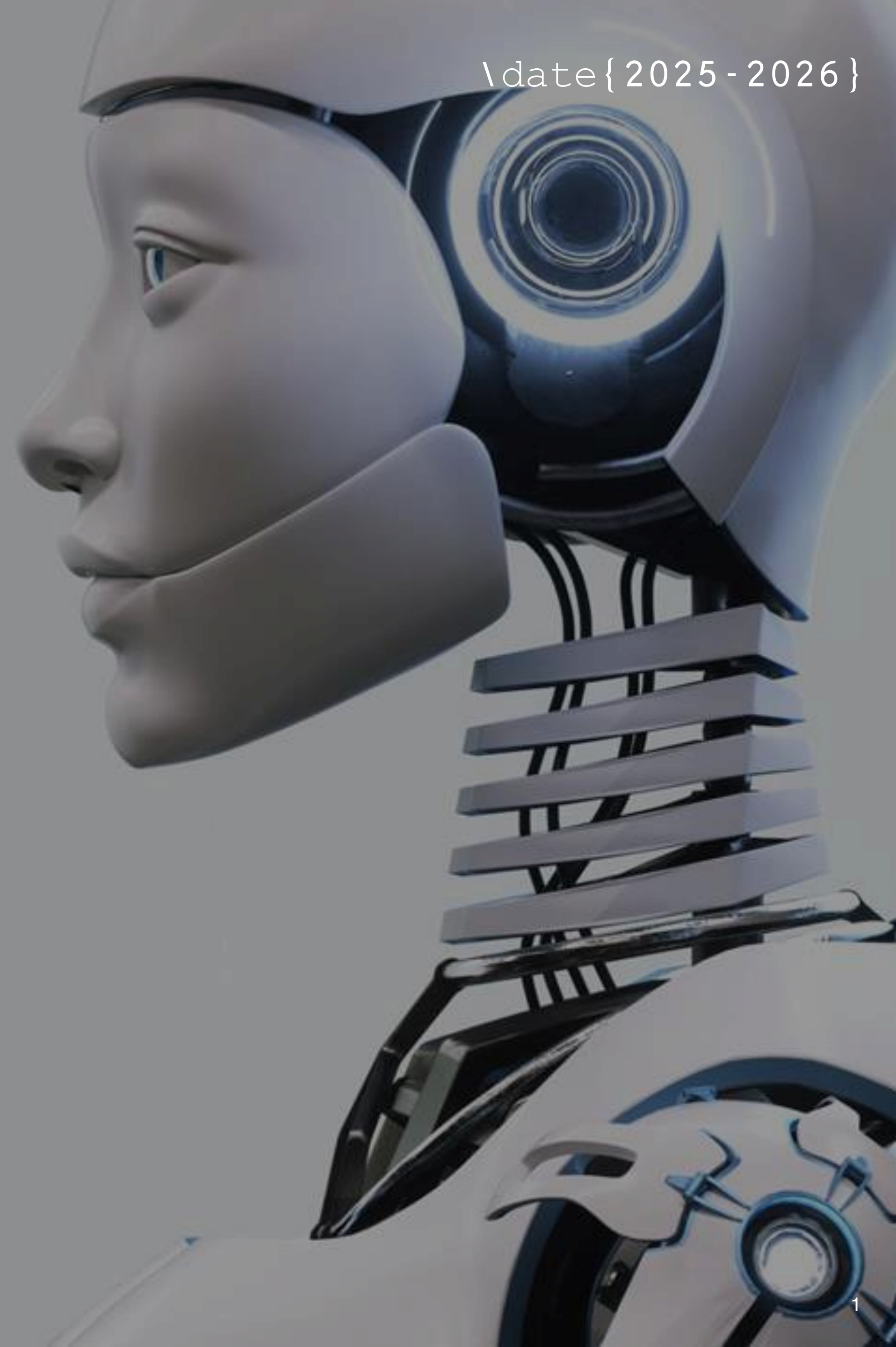

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ABOUT ME

Here I present my academic and professional experience, as well as a short overview of my skills.

Education : Overview

- **2024 – Present: Sorbonne Université & ENSAM (Paris)**
  Master's Degree in Robotics (Year 2, Joint Program with ENSAM) + CMI
- Specializations: Mobile Robotics, Haptics, Simulation, Artificial Intelligence for Robotics
- Core modules: Robotic Manipulators, Modelling and Control

- **2024 – National Taiwan University (Taipei)**
- mechanical systems and numerical methods

- **2020 – 2023: Sorbonne Université (Paris)**
  Bachelor's Degree in Mechanical Engineering + CMI
- Focus on mechanical systems, dynamics, and applied mathematics

Education : CMI

During this 5 years I also followed the CMI Mechanics Program at Sorbonne Université

The *Cursus Master en Ingénierie (CMI)* is a **five-year selective program** built on top of a Bachelor's degree and a Master's degree.

Unlike a standard curriculum, the CMI requires:

- **More coursework**: **36** ECTS / semester instead **30**
- **Additional subjects** such as *innovation, entrepreneurship, project management, and more English course*.
- **Professional experience**: mandatory internships
- **International mobility**: a required study abroad

The program is designed to train students not only as engineers but also as future **innovators and researchers**.

Professional Experience:

I had to work during my academic studies, so I gained some professional experience through student jobs and summer jobs.

1. I worked as a production worker in a food factory, as a security guard, and as a cleaning agent during the summer.

2. During this academic year, I also worked as a cleaning agent.

Skills Overview

Development Tools 💻

- Python
- C,C++
- Matlab
- LaTeX
- HTML/CSS
- Arduino

Technical Skills ⚙️

- ROS2, Gazebo, RVIZ
- SolidWorks
- 3D Printing
- Docker / Virtual Machine
- OpenCV
- Fritzing
- Raspberry Pi, Arduino
- GitHub
- OOP (Object-Oriented Programming)
- RDM6
- Unity (in progress)

Languages

- French (Fluent)
- English (Intermediate)
- Hebrew (Beginner)

Certifications

During my academic progress I had to pass some certifications like :

1. MOOC: Brand Management by University of London and London Business School (in progress)

2. TOEIC : score (in progress)

3. License B

Project Summaries

In this following slides I noticed some important project I did during my academic and personal experience in chronological order.

For more information, I will glad to see you on my [GitHub:Dan-project].

Academic Project : Artificial Canal (1Yr)

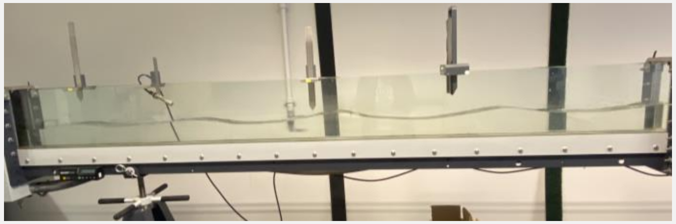


Figure : Artificial canal

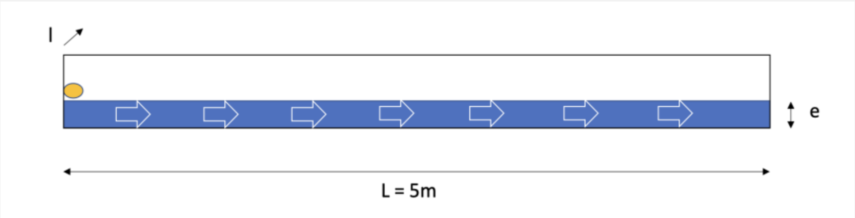


Figure : Plan of the system

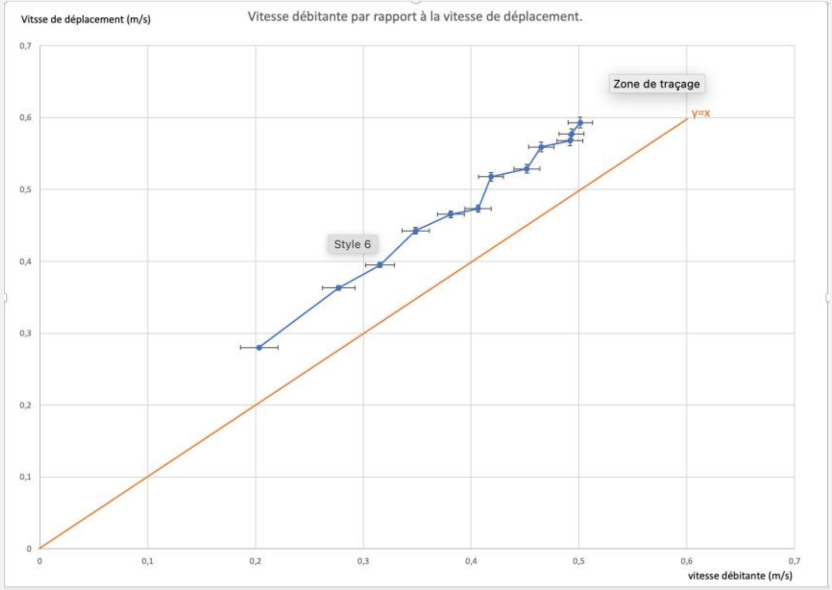
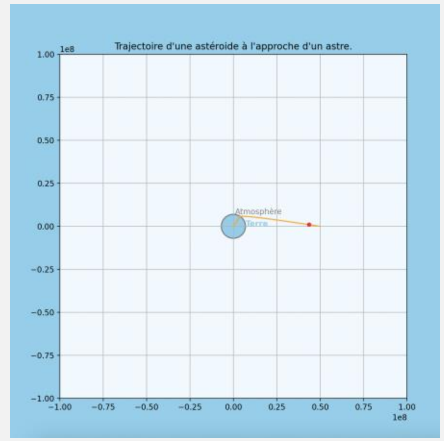


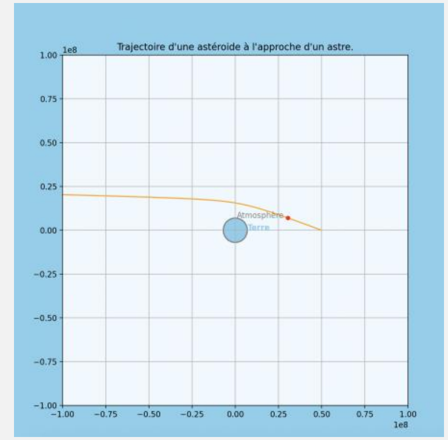
Figure : flow (orange) vs velocity of the float

- First step into a mechanical project
- All measurements include uncertainty.
- Type A: from statistical variation in repeated trials.
- Type B: from external sources (instrument specs, calibration, references).

Academic Project : Study of Meteorite approaching Earth (2yr)

Simulation of a meteorite's trajectory under Earth's gravity and aerodynamic drag.





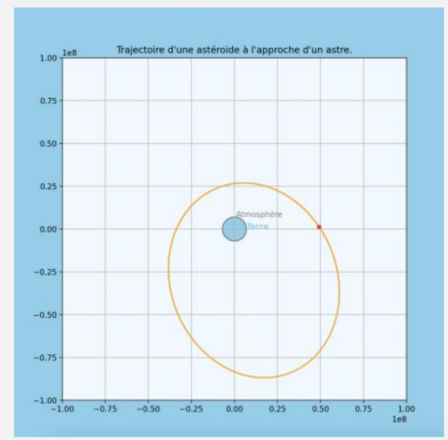


Figure : same velocity diffenrent angle

Key elements

- Python (Matplotlib), C
- **Forces:** Earth's gravitational attraction and aerodynamic drag
- **Important parameters:** initial angle, initial speed, drag coefficient...
- **Fundamental principles:** Newtonian dynamics — second law F=ma.
- **Numerical method:** Verlet integration to update positions and velocities at each time step.

Academic Project : OpenRov (2 yr)

OpenRov was a marine robotics company focused on democratizing underwater exploration.

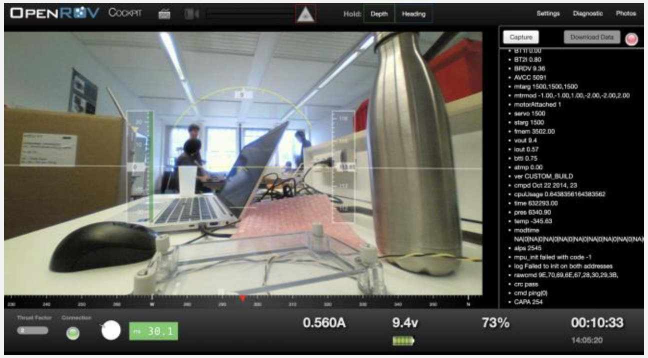


Figure : OpenRov Cockpit

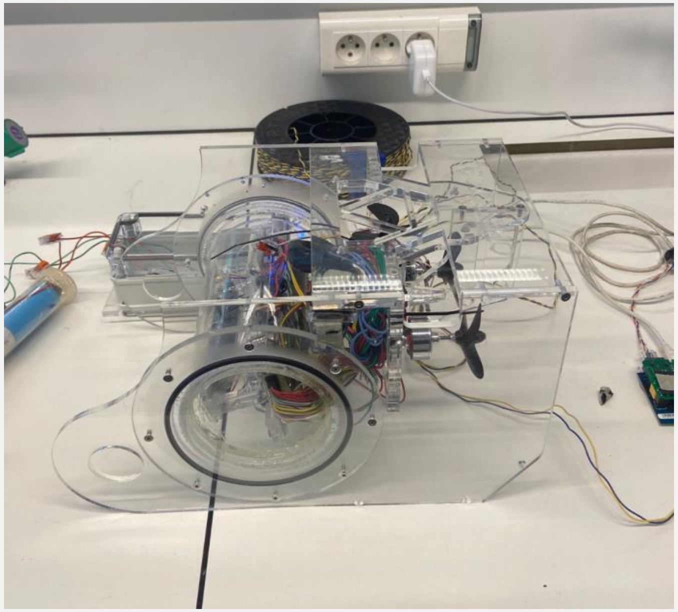


Figure : OpenRov

Missions:

- Install ultrasonic sensors for autonomous operation (implementation of SLAM).
- Work on the waterproofing of the submarine.
- Integrate the IMU and the pressure sensor.
- Research on the sorftware part

Academic Project : Video Games(3 yr and 4 yr)

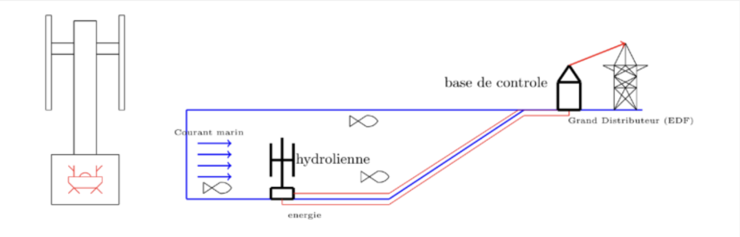
During my academic experience, I had to work on two similiar project : create a video games with Python with using OOP (oriented object programming).

These projects helps me to learn how to structure a code and share with my teamates on GitHub.

Figure : Tank Games

Figure : Marvel Games

Academic Project : Hydroflow (4yr)



Schema in/out

In our project management course, we worked in a team to develop a start-up concept, covering all key preliminary steps such as :

- Context analysis
- Macro risks
- Planning
- Project risks
- Needs → functions → macro system
- Structuring
- Budgeting
- Profitability study

The course was supervised by instructors from **Safran**.

Risques	Probabilité	Impact	Niveau de risque	Actions
Défaillance des équipements	Élevée	Critique	Élevé	Maintenance & choix de matériaux
Environnementaux	Modérée	Critique	Élevé	Suivi météo & études environnementales
Installation	Élevée	Critique	Élevé	Technologies autonomes
Accidents de travail	Faible	Critique	Moyen	Formation & protocoles
Modification des courants	Faible	Modéré	Faible	Études hydrodynamiques
Sociaux	Faible	Modéré	Faible	Communication & consultation

Figure: Macro risks

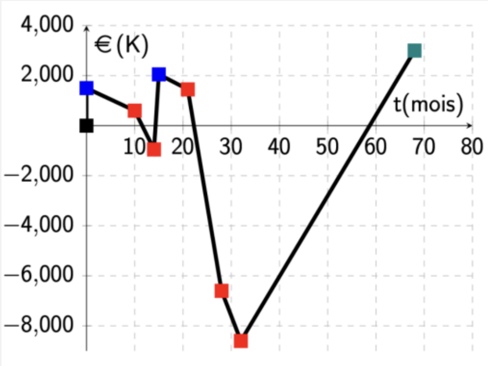


Figure: Budgeting

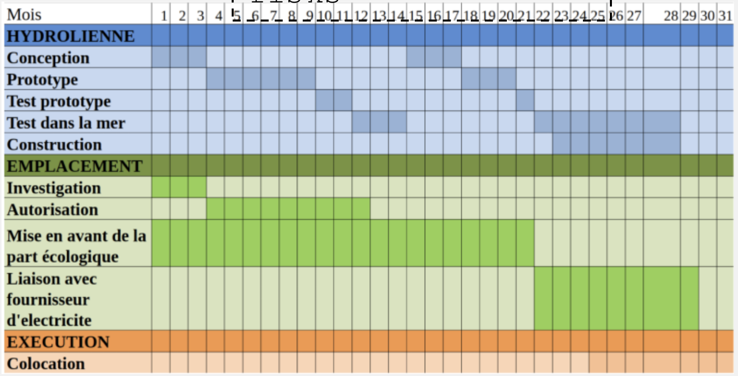


Figure: Planning

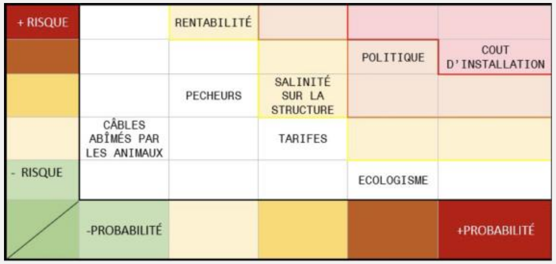


Figure: Risks

Academic Project : 3R Parallel Robot(4yr)

The objective of this project was to design a 3RRR-type manipulator robot model in SolidWorks and to simulate this model using Python.

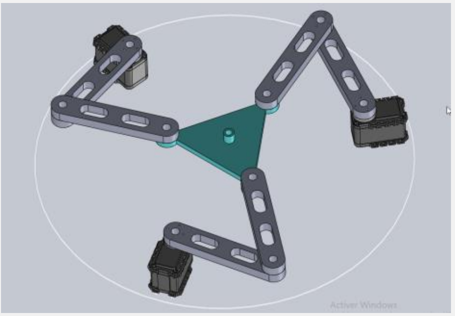


Figure: Conception 3D

- Direct and Inverse Geometric Modeling
- Kinematic Modeling to Identify Singularities

$$A v_i = B \dot{\alpha}_i \qquad (1)$$

$$\begin{pmatrix} \cos\gamma_1 & \sin\gamma_1 & d_1 \\ \cos\gamma_2 & \sin\gamma_2 & d_2 \\ \cos\gamma_3 & \sin\gamma_3 & d_3 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \\ \omega_z \end{pmatrix} = \begin{pmatrix} e_1 & 0 & 0 \\ 0 & e_2 & 0 \\ 0 & 0 & e_3 \end{pmatrix} \begin{pmatrix} \dot{\alpha}_1 \\ \dot{\alpha}_2 \\ \dot{\alpha}_3 \end{pmatrix}$$

- Simulation in Python: Identification and Avoidance of Singularities
- SolidWorks : Identifying the Geometric Parameters of the Optimal Robot

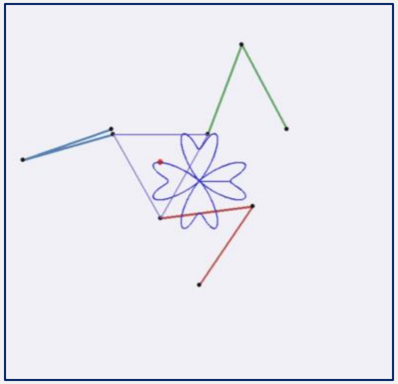


Figure: Trajectory Tracking

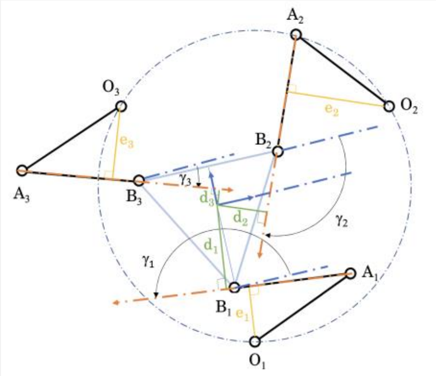


Figure: Kinematic model and parameter

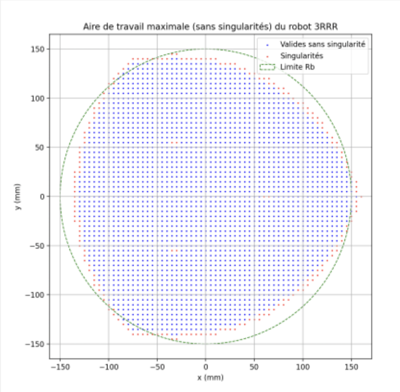


Figure: Theoretical Optimal Surface

Academic Project :Onboard Storage Design Project for a Tram (4yr)

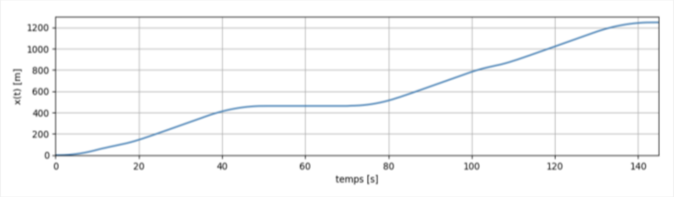


Figure: train position over time

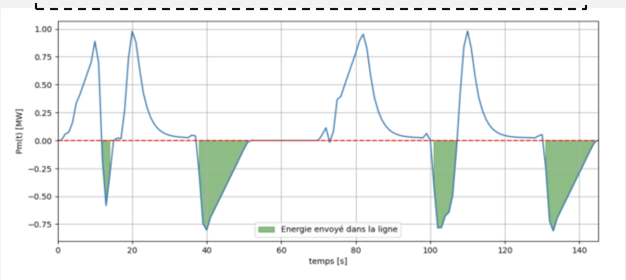


Figure: power of train over time

- **Objective:**Optimize **voltage drop (ΔV)** and **battery capacity** for an onboard energy storage system in a tramway.

- **Methodology:**
- System modeling with and without battery.
- Energy management (charging/discharging).
- Optimization through Monte Carlo simulation and Genetic Algorithm (NSGA-II).
- interpretation of the Pareto front.

- **Results:**
- Optimal battery capacity: **4,500 – 21,500** kWh.
- Operating threshold: **245 kW – 550** kW.
- Reduced voltage drops and improved grid stability.

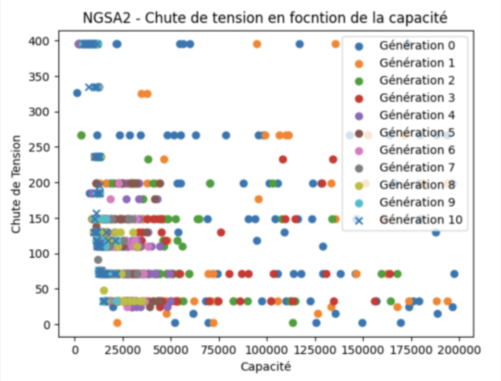


Figure: Genetic algorithm

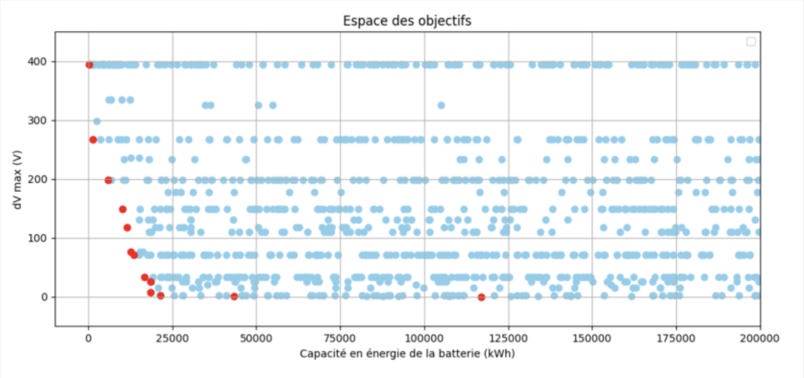


Figure: Monte-Carlo (red: Pareto front)

Academic Project : Turtlebot ROS2(4yr)

- Connect to the robot with
- Follow a path between two lines.
- Use **OpenCV** with the camera to calculate the barycenter of each line and anticipate turns.
- Implement a decision system for roundabouts (choose left or right exit).
- Integrate **obstacle avoidance with the LIDAR**.
- Navigate through **corridors wwith the LIDAR**.
- Place a ball into a goal.
- Test first in **Gazebo simulation**, then deploy on the real robot.

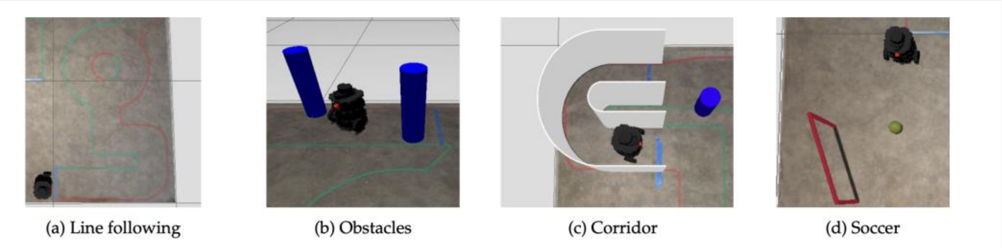


Figure: Tasks

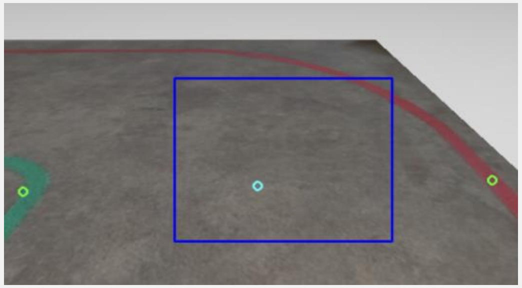


Figure: Camera on Gazebo (simulation)

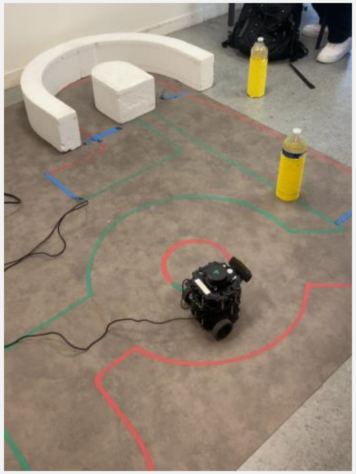


Figure: Real route

Personal Project : Home automation

- Built a **custom smart home automation system** with Home Assistant on Raspberry Pi
- Designed case on Solidworks with internal ventilation
- Integrated **ESP32 microcontroller** to control LED lighting
- Combined **hardware, software, and IoT integration** for a seamless solution

Internship Experience

Here I will notice all my intership experience I had during this 4 years.

Structural Reinforcement: EMC TECHNI & CETO (2sd yr)

Worker internship in the field of Structural Reinforcement (1 month)

- Worker Internship in the field of Structural Reinforcement Tremblay-en-France, France
- – Site preparation.
- – Building masonry walls, making concrete.

- Destructive testing (e.g., locating piping and buried basements).
- – Installation of reinforcement beams.

Figure: Sondage destructif

Figure: Livraison

Figure: Coffret chantier

Figure: Évacuation débris

Structural Reinforcement: EMC TECHNI & CETO (2sd yr)

Study Office Internship in the field of Structural Reinforcement and Diagnostics (1 month)

- On-site intervention: Structural diagnostics (destructive and non-destructive testing) and assisting in work management.
- Assistant works manager (guide the excavator, draw up plans)

Figure: Non-destructive testing

Figure: G1 Gauge installation

Figure: Measure the deph

Figure: destructive testing

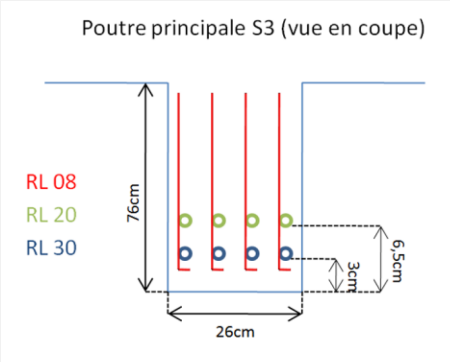


Figure: Schema of the beam

Structural Reinforcement: EMC TECHNI & CETO (2sd yr)

- Establish reinforcement plans using calculations at **SLS** and **ULS**, performed with the software **RDM6**.

- Report writing to recommend reinforcements.

Données	Valeurs
Chargement linéique sur la poutre (kN / ml : ELS)	5,9
Longueur de la poutre : L (m)	4,9
Poids linéique de la poutre (daN)	43,5
Coefficient de calcul de flèche : k (-)	350

Données recherchées	Résultats
Module de Young : E (GPa)	210
Inertie de la poutre : I (cm^4)	2492
Flèche maximale : f_max (mm)	14
Flèche calculée (mm)	9

Figure: Example of calculation. of SLS

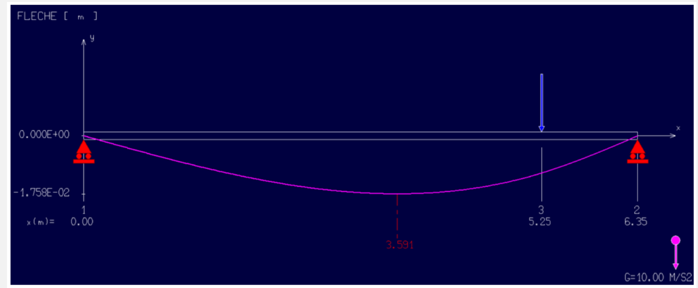


Figure: Simulation of Flexion beam on RDM6

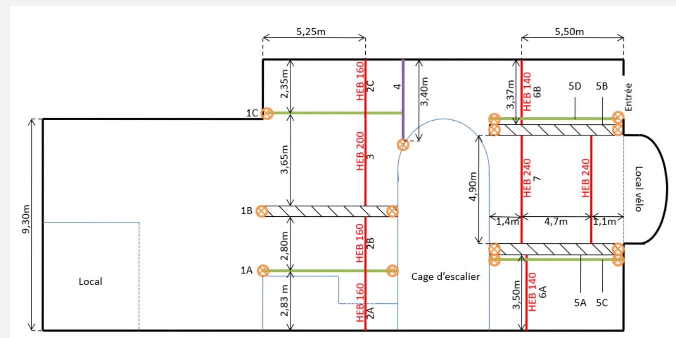


Figure: Measure the deph

SLS (Serviceability Limit State) – ensures that the structure performs properly under normal service conditions (no excessive deflection, cracking, or vibration). It’s about *comfort and durability*.

ULS (Ultimate Limit State) – checks that the structure can safely resist maximum loads without collapsing or losing stability. It’s about *safety and strength*.

Propagation of acoustic waves in phononic crystals (Institut Jean le rond d'alemebert)

A **phononic crystal** is an artificial periodic structure in which **acoustic waves** propagate.

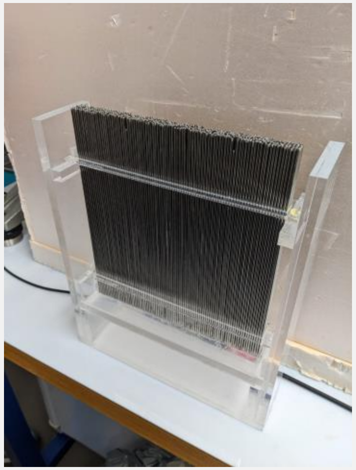


Figure: Phononic crystals

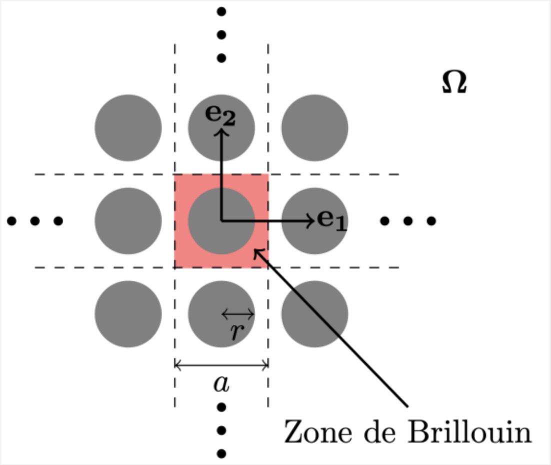


Figure: Brillouin aera

Alembert equation

$$\partial_{tt}u - c_0^2\left[\partial_{xx}u + \partial_{yy}u\right] = 0$$

General Solution of alembert equation

$$u(x,t) = F(x - c_0 t) + G(x + c_0 t)$$

Equation in crystals
(non penetration condition)

$$\begin{cases} \mathbf{v}\cdot\mathbf{n} & = 0 \quad \text{sur} \quad \partial_n\tilde{\Omega} \\ \partial_{tt}u(\mathbf{x},t) - c_0^2\Delta u(\mathbf{x},t) & = 0 \quad \text{dans} \quad \mathbb{R}^2 \end{cases}$$

Dispersion relation

$$\omega_n(k) = c_0\sqrt{\lambda_n(k)} \Rightarrow f_n(k) = \frac{c_0}{2\pi}\sqrt{\lambda_n(k)}$$

Propagation of acoustic waves in phononic crystals (Institut Jean le rond d'alemebert)

Experimental Method

We send waves of a certain frequency band into the water (better wave propagation) through our crystal (at a chosen angle).

Then we had to process the obtained data (gate functions, FFT) to erase the noise and the data outside of the band wave.

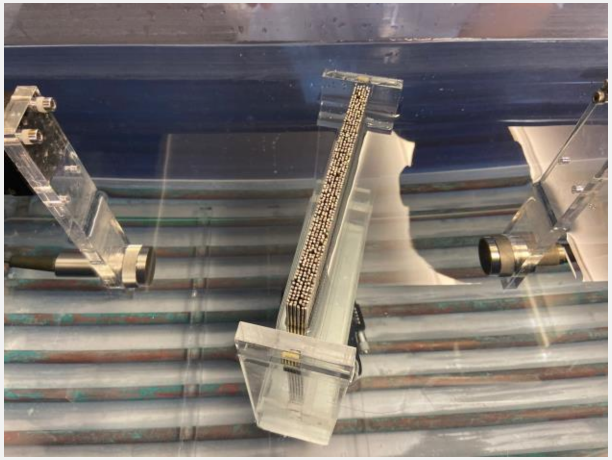


Figure : Experience measurement

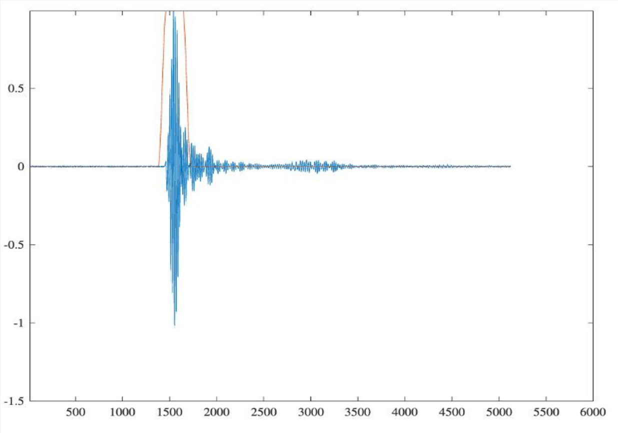


Figure : data (blue) and gate (orange)

Propagation of acoustic waves in phononic crystals (Institut Jean le rond d'alemebert)

For certain frequencies, waves do not pass through the crystal.
These frequency zones are called band gaps.

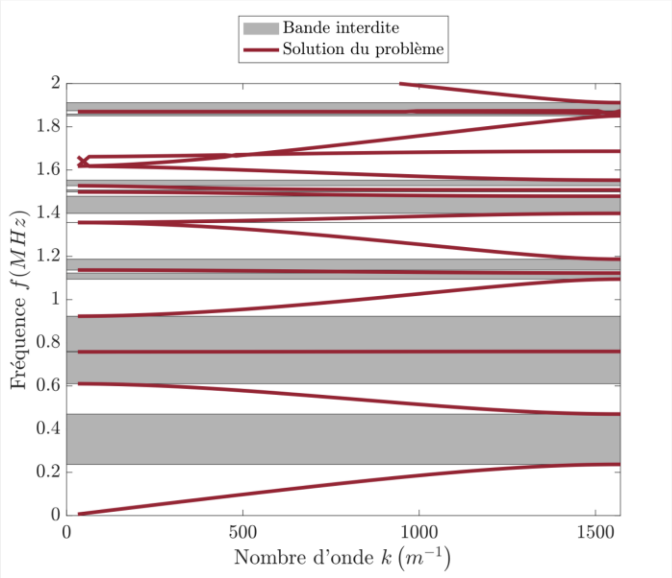


Figure :Theoretical solution

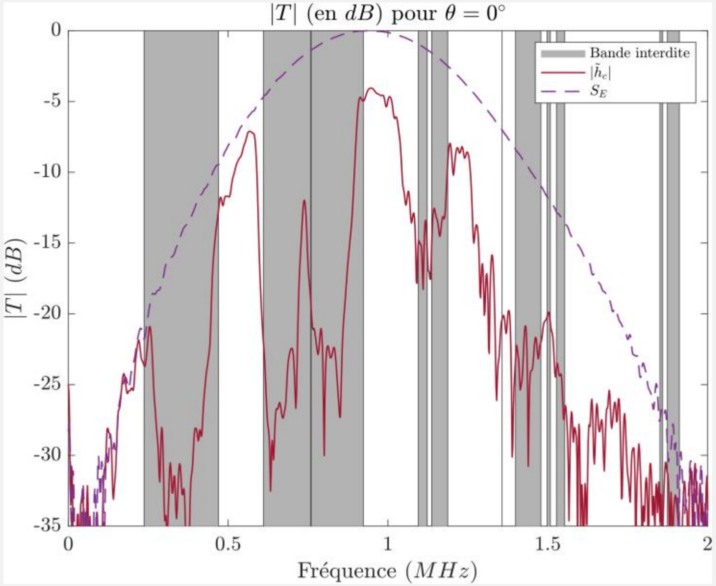


Figure : Experimental and Theoretical (grey band) solution

We did this with different angles of incidence of the wave and we saw that the forbidden bands differ according to the angles also

Development of robotic equipment in the master's department

CONTEXT: Provide solutions to the difficulties encountered by students during two practical robotics projects.

Robot Pincher PX100

Challenges:

- The installation guides were not suitable for macOS users (since Docker Desktop does not allow USB access), and some installation issues could occur on Windows.

Objectives:

- Multiplatform deployment of the Pincher robot
- Installation and configuration guide

Robot RRR Plan serial

Challenges:

- **The MATLAB-based lab sessions** were being used less and less by students, in favor of **Python**, which was considered more accessible. Several **robots** experienced **hardware failures** or **connection issues** with the **Dynamixel motors**.

Objectives:

- **Modernize** the **3R robot simulation code** (from **MATLAB to Python**)
- **Repair and organize** the robots
- **Adapt and update** the **robotics lab sessions**

Development of robotic equipment in the master’s department

Methodology and Results:

- Study of Linux simulation methods and their advantages/disadvantages
- Virtualization using an image compatible with the host architecture and ROS Interbotix
- Installation of packages adapted to each architecture
- Installation guide covering most use cases
- Testing on different machines for validation

- Python simulation using OOP (Object-Oriented Programming)
- Application of direct and inverse geometric models
- Control via Dynamixel-controller
- Adaptation of lab sessions for compatibility with the new code
- Repair and preparation of the robots

Figure: Simulation (initial and final position)

Cooperative multi-robot semantic mapping via LiDAR-camera fusion

- Algorithm development (LIO-SAM & YOLO): LiDAR-camera fusion (3D LiDAR + 2D Color Camera) using YOLO semantic segmentation and LIO-SAM with loop closure to build 3D semantic maps.

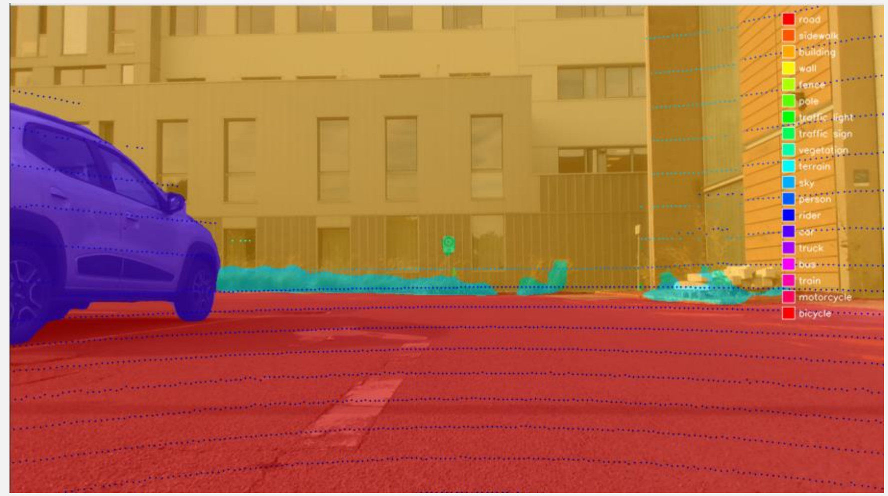


Figure : Lidar points + Semantic masks

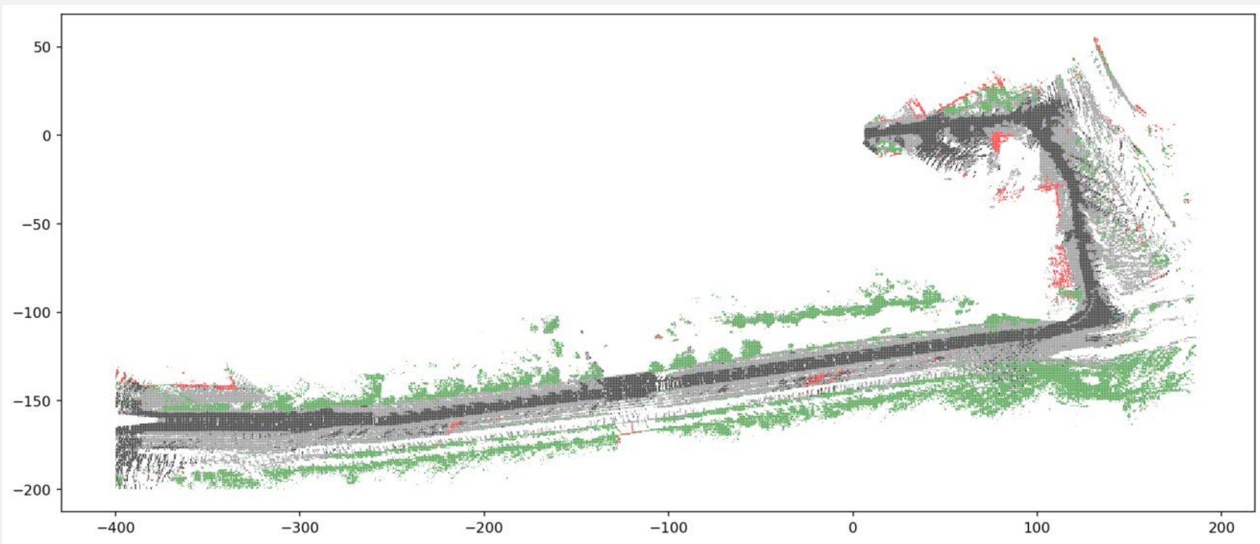


Figure : Map with Kitti Dataset

Dataset validation: Algorithm training, benchmarking, and initial validation on the KITTI dataset.

Cooperative multi-robot semantic mapping via LiDAR-camera fusion

- Map *a-priori* generation from OSM: Creation of *a priori* maps for semantic mapping contextualization and autonomous navigation planning.

Figure : OSM Map

Figure : Discretized map

Cooperative multi-robot semantic mapping via LiDAR-camera fusion

- Real-world robot testing & data collection: Robot teleoperation and ROS bag creation on physical AgileX Ranger and Scout Mini platforms.
- System validation: Standalone testing and validation using the pre-built *a priori* map.

Figure : Data acquisition

Multi-robot extension: Scaling the architecture to a cooperative multi-robot system with distributed map merging and inter-robot loop closure.

Goals and aspirations

I am completing a Master’s in Advanced Systems and Robotics and aim to pursue either an industrial PhD or a career in robotics engineering.
I’m particularly interested in working on mobile and humanoid robots, focusing on their autonomy, perception, and control.

Contact:

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